

A Perceptual Extension of Skin-Stretch Sensation on the Back Using Vibrotactile Cues

Kenta Tanizawa^{1,2}, Yugo Nakamura¹, and Shogo Fukushima¹

¹ Kyushu University

² JSPS Research Fellow

Abstract. Skin-stretch stimulation provides clear directionality and is readily perceived as a deformation sensation; however, its implementation—particularly in wearable devices—is strongly constrained by actuator count and mechanical structures. The back has attracted attention as a tactile display site due to its large surface area, while humans tend to integrate ambiguous and sparse tactile information rather than precisely localizing individual stimuli. In this study, we exploit this perceptual characteristic by focusing on the back, where tactile receptors are sparsely distributed, with the aim of modulating the perceived skin-stretch sensation. We designed a wearable device that presents vibration stimuli around the skin-stretch location on the back and investigated whether the addition of vibration shifts the perceived onset position of skin-stretch. Our results indicate that vibration can effectively alter the perceived traction location, suggesting a means to transcend mechanical constraints inherent in conventional wearable devices, such as action–reaction limitations. This work provides design insights for future tactile displays that leverage human perceptual characteristics.

Keywords: Skin-stretch · Tactile Integration · Dorsal Surface.

1 Introduction

Aristotle classified human sensation into five modalities as a framework to explain the functions of the animal soul. Among these, tactile sensation is understood as a form of mechanoreception and is regarded as the most primitive sensory modality, acquired early in evolution for perceiving the external environment.

Among haptic feedback techniques, skin-stretch (tangential skin deformation) cannot be simply replaced by commonly used vibration-based feedback, which primarily conveys sensations related to intensity rather than direction or deformation[10][6]. So, a wide range of devices have been developed to deliver specific forms of skin-traction[3][2][16][13]. However, implementing skin-stretch-based feedback in a wearable form factor involves design constraints that differ fundamentally from those of vibrotactile systems[12]. Vibrotactile systems typically adopt an approach in which many small actuators are distributed across a surface[19][11][9]. In contrast, scaling skin-stretch to high spatial density is

structurally challenging. This is because each stretch point requires mechanical actuation to generate tangential displacement, making it difficult to satisfy the size and weight constraints demanded of wearable devices. In addition, due to their wearable nature, skin-stretch devices are mounted on the body, which results in tangential forces being applied simultaneously to two or more contact points through action–reaction. These forces may be perceived as unintended tactile stimuli.

A considerable body of research has explored perceptual approaches to providing diverse haptic sensations without relying on complex mechanical mechanisms. For example, the cutaneous rabbit illusion[5] and referred thermal sensations[7] demonstrate that discretely presented stimuli can be perceived as continuous motion across the skin, enabling the generation of continuous tactile experiences with a small number of actuators. These phenomena are not merely illusions but suggest that the human tactile system is capable of constructing meaningful sensations from sparse and ambiguous inputs. Furthermore, studies have reported that combining different types of tactile stimuli can modulate the perceived sensation itself. Examples include the pulling illusion[17], in which asymmetric vibration induces a sensation of being pulled despite the absence of actual external force, and phenomena in which rapid changes in thermal stimulation elicit sensations of pressure or traction[18]. Collectively, these studies indicate that the integration of heterogeneous tactile inputs can give rise to sensations not directly encoded by the corresponding receptors, suggesting that vibrotactile stimuli can function not only as auxiliary cues but also as signals that complement and extend the perception of deformation and force.

Meanwhile, the back as a site for haptic stimulation has primarily attracted attention in the context of body extension and large-area tactile displays[16][4]. However, it is characterized by a low density of tactile receptors and limited opportunities for self-touch or visual reference in everyday life[1]. These properties suggest that the body representation of the back is relatively coarse and that sparse or ambiguous tactile information may be more readily integrated, rather than individual stimuli being perceived with precise spatial localization.

In this study, we focus on these characteristics of tactile perception and investigate whether combining skin-traction and vibrotactile stimulation can spatially extend the perceived sensation of skin-stretch on the back beyond the physically stimulated locations. Through this approach, we aim to overcome the limited scalability of spatial density in skin-stretch interfaces as well as unintended stimuli arising from action–reaction effects. Unlike prior tactile illusion studies such as the cutaneous rabbit effect or apparent motion, which primarily interpolate discrete stimuli to create a sense of continuous movement, this work focuses on the modulation of perceived deformation origin in skin-stretch feedback. Specifically, we demonstrate that vibrotactile cues can systematically shift the perceived onset location of a directional deformation signal (skin-stretch), rather than merely generating interpolated motion or intensity-based sensations. This distinction is critical for wearable haptic interfaces, where conveying directional force and deformation is essential but mechanically constrained.

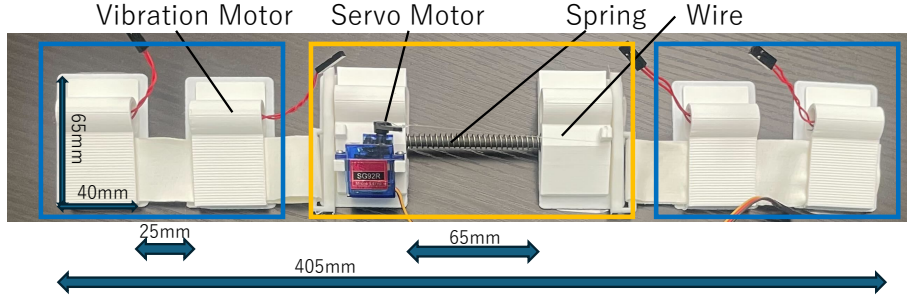


Fig. 1: Overview of the wearable skin-traction and vibrotactile device. Components enclosed in yellow brackets correspond to the skin-traction mechanism, whereas components enclosed in blue brackets correspond to the vibrotactile stimulation mechanism.

The contributions of this work are summarized as follows:

- The integration of skin-traction and vibrotactile stimuli on the back is experimentally shown to induce an extension and a shift in the perceived location of skin-stretch sensation.
- The effects of the spatial arrangement of vibrotactile stimuli on the perceptual integration of skin-stretch sensation are clarified.
- Design implications are provided for constructing higher-level deformation sensations in haptic displays under constraints on the number of available actuators.

2 Device Design

This study aims to develop a haptic feedback system as a wearable device. The external appearance of the device is shown in Fig. 1. This device is composed of six parts: two central components used to perform skin-traction, and four outer components that provide vibration. All of these are secured with biocompatible fabric. These components are secured to the body using a biocompatible fabric.

The total mass of the device is 150 g, and it is attached near the waist using a biocompatible adhesive gel (SIXPAD, MTG Co., Ltd., Japan; 37 mm × 64 mm; acrylic-based conductive hydrogel). The adhesive area measures 37 mm × 64 mm. The minimum distance between the two skin-traction components is 65 mm, while the minimum distance between the other components is 25 mm, resulting in a total device length of 405 mm. Under these spatial conditions, adjacent components are located within a range comparable to reported two-point discrimination thresholds, whereas components separated by one intervening component are positioned outside the two-point discrimination threshold [8].

Skin-traction stimulation is achieved using servo motors (SG90, TowerPro) in combination with guiding mechanisms that constrain the direction of motion, allowing the adhesive contact area on the skin to be precisely displaced

in the horizontal direction. Vibrotactile stimulation was provided using an eccentric rotating mass (ERM) vibration motor. The motor has dimensions of 28×12 mm, an operating voltage of DC 3–6 V, a nominal voltage of 3 V, and a current consumption of approximately 32 mA. Based on the motor’s rotational speed observed when 5 V was applied, the vibration frequency was calculated to be approximately 250 Hz. This vibration frequency was selected based on the anatomical distribution of mechanoreceptors in hairy skin, where Pacinian corpuscles are more densely distributed than Meissner corpuscles.

Skin-traction is generated using servo motors, enabling a maximum displacement of 35 mm (17.5 mm in each direction). Because the device’s gel is completely fixed to the skin, it enables precise traction in the shear direction along with the device’s deformation, without applying force perpendicular to the skin. Regarding the spatial configuration, a vibrator was placed 25 mm away from the skin-traction location toward the abdomen. Experiments were conducted in which vibration and skin-traction were applied simultaneously. When vibration was applied at a location 25 mm away from the skin-traction site, changes in the perceived skin-stretch sensation were observed. In contrast, no such changes were observed when vibration was applied to the abdomen simultaneously with skin-traction. These effects were verified through experiments conducted with five participants.

To minimize the perceived weight of the device, particular attention was paid to reducing the overall mass. As a result, the design prioritizes a lightweight structure, and the total weight of the device is limited to 150 g, thereby preventing a decrease in user acceptance due to excessive weight[14]. Furthermore, user acceptance is improved by avoiding constrictive attachment methods such as belt-type fasteners[15]. These design considerations aim to enhance user comfort and improve overall usability.

The appearance of the device when worn is shown in Fig. 2. Each device is arranged to extend from the back toward the lateral abdominal region, straddling the spine. Skin-traction stimulation is applied to relatively flat regions of the back near the spine, while vibration actuators are attached to regions with greater surface curvature. The vertical placement of the device is approximately 40 mm above the navel. This position was selected because skin deformation caused by trunk rotation and limb movements is relatively small in this area, thereby reducing discomfort during wear and improving the overall wearing experience of the wearable device.

3 Experiment

3.1 Purpose and Research Questions

The purpose of this experiment was to investigate how the perceived location of skin-stretch sensation changes when vibrotactile stimulation is applied simultaneously with skin-traction, and to obtain design-relevant insights for wearable skin-stretch displays.

Specifically, this study addressed the following research questions (RQs):

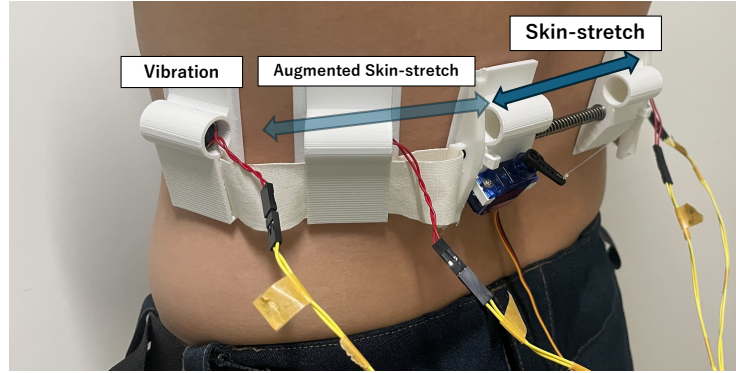


Fig. 2: The device mounted on the participant's back.

- **RQ1:** Does simultaneous vibrotactile stimulation modulate the perceived location of skin-stretch sensation beyond the physically actuated site?
- **RQ2:** Is the modulation of perceived skin-stretch sensation spatially symmetric when vibrotactile stimuli are applied symmetrically?
- **RQ3:** How does the spatial sparsity of vibrotactile actuator placement affect the perceived extension of skin-stretch sensation?

To address these questions, we conducted a controlled laboratory experiment in which multiple patterns of vibrotactile stimulation were presented simultaneously with skin-traction on the back, and participants reported the perceived location of the skin-stretch sensation.

3.2 Participants

Twelve participants (ten males and two female; age: 22.58 years on average, $SD = 1.08$, range: 21–25) took part in the experiment. All participants were healthy adults with no reported sensory impairments.

This study was approved by the institutional ethics review board, and informed consent was obtained from all participants prior to the experiment.

3.3 Experimental Design

A within-participant experimental design was employed. Each participant experienced all stimulus conditions.

Eight stimulus conditions were prepared by combining one skin-traction unit with different activation patterns of four vibrotactile actuators arranged horizontally on the back. Table 1 summarizes the activation patterns of the skin-traction unit and vibrotactile actuators in each condition.

Condition 1 served as the baseline condition, in which only skin-traction was applied without vibration. Conditions 2 and 4, as well as Conditions 3 and 5, were

Table 1: Activation of vibrators and skin-traction units in each condition.

Condition	Leftmost vibrator	Second-left vibrator	skin-traction	Second-right vibrator	Rightmost vibrator
1	×	×	○	×	×
2	×	○	○	×	×
3	○	○	○	×	×
4	×	×	○	○	×
5	×	×	○	○	○
6	○	○	○	○	○
7	○	○	○	○	×
8	○	×	○	×	×

designed as left–right symmetric pairs to examine whether vibrotactile stimulation induces symmetric modulation of perceived skin-stretch sensation. Conditions 6 and 7 were included to investigate perceptual effects when vibration was applied on both sides simultaneously. Condition 8 was introduced to examine the effect of sparse vibrotactile actuator placement, which is relevant to practical device design.

Condition 1 was presented first as a reference to anchor participants’ perception of the baseline skin-stretch location. We acknowledge that this may introduce a potential order effect; however, we prioritized establishing a consistent perceptual reference. The remaining conditions were counterbalanced using a Latin square design.

3.4 Stimuli and Apparatus

All stimuli were initiated simultaneously. Skin-traction was applied bilaterally at a precisely controlled velocity of 350 mm/s using a servo motor, resulting in a skin stretch of 35 mm. After 0.5 s, the skin was returned to its original position at the same velocity. Thus, the total duration from the onset to the end of skin-traction was 0.7 s.

Vibrotactile stimulation was applied by supplying a 5 V voltage to the vibration motor, producing a frequency of approximately 250 Hz for 0.6 s. The onset of vibrotactile stimulation was synchronized with the onset of skin-traction.

In other words, skin-traction was applied over 0.1 s, and vibrotactile stimulation was presented simultaneously. The deformed skin state was then maintained for 0.5 s, after which the skin was returned to its original position over 0.1 s, at which point the vibration was also terminated.

3.5 Procedure and Measures

For each trial, participants were instructed to focus on the sensation of skin stretch perceived on their back. After each stimulus presentation, participants reported the perceived location of the skin-stretch sensation by marking the position where they felt the skin was being stretched on a schematic diagram of the back. This method relies on subjective spatial reporting and may introduce variability due to differences in body representation and drawing accuracy.

In addition, participants rated their confidence in their perceived location using a seven-point Likert scale (1: not confident at all, 7: very confident). These two measures were used to capture both the spatial characteristics of perceived skin-stretch sensation and the subjective reliability of the perception.

4 Results

4.1 Changes in perceived traction induced by actuator-driven vibration

Figure 3 summarizes, for each condition, the distance from the center of the back (spine) to the location where the onset of skin-traction was perceived. Here, the distances were normalized by setting the distance from the center of the back to the most lateral point on either side (the lateral abdomen) to 1. Table 2 summarizes the mean and standard deviation for each condition, reported separately for the left and right sides.

In this study, stimuli under eight conditions were presented to twelve participants, and response values were obtained separately for the left and right sides. All factors were treated as within-subject variables. The normality of the data for each condition was examined using the Shapiro–Wilk test. As a result, normality was not rejected for any condition (all $p > .05$).

A one-way repeated-measures analysis of variance (ANOVA) was conducted separately for the left and right sides, with condition (eight levels) as the within-subject factor. When a significant main effect was observed, post hoc comparisons were performed using paired-samples t -tests with Holm correction. As measures of effect size, partial eta squared (η_p^2) was calculated for the ANOVA, and Cohen’s d_z was calculated for the post hoc tests.

For the left side, the results of the one-way repeated-measures ANOVA revealed a significant main effect of condition ($F(7, 77) = 32.42$, $p < .001$, $\eta_p^2 = .75$). Post hoc comparisons showed that Condition 1 differed significantly from Conditions 2, 3, 6, 7, and 8. In addition, Condition 2 showed significant differences from Conditions 3, 4, 5, 6, and 7, and Condition 3 differed significantly from Conditions 4 and 5 (Holm-corrected $p < .05$). The effect sizes for all significant comparisons were large, with Cohen’s d_z ranging from 1.12 to 2.56.

For the right side, a one-way repeated-measures ANOVA also revealed a significant main effect of condition ($F(7, 77) = 31.98$, $p < .001$, $\eta_p^2 = .74$).

In addition, paired equivalence testing was conducted separately for the left and right panels using the Two One-Sided Tests (TOST) procedure (Table 4). Only pairs for which equivalence was supported at a significance level of $\alpha = 0.05$ are reported. For each pair, the mean difference and the corresponding TOST p -value are presented. The equivalence margin was set to $\Delta = 0.1$. This value approximately corresponds to the spatial resolution defined by the actuator spacing (25 mm ≈ 0.11 in the normalized scale). This choice reflects the smallest spatial difference that can be meaningfully distinguished based on the physical configuration of the device. The significance level was set to $\alpha = 0.05$, and equivalence

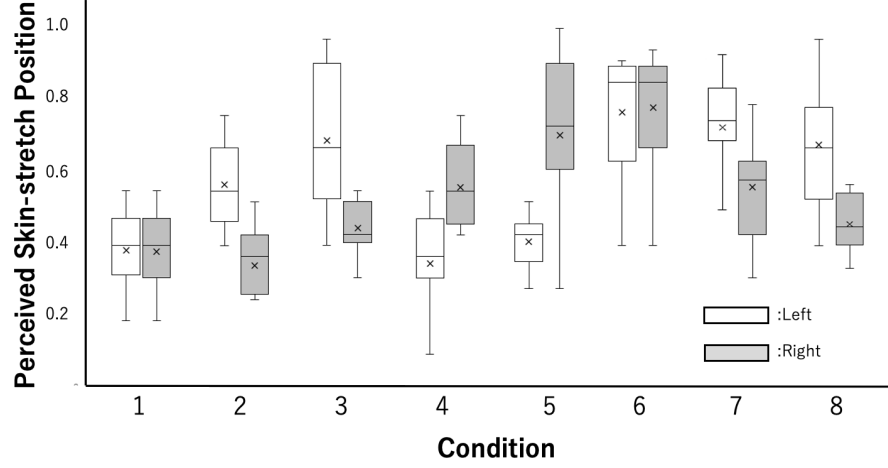


Fig. 3: Perceived onset location of skin-stretch under each stimulus condition. The normalized distance from the center of the back (spine) is shown for the left and right sides. Each boxplot summarizes participant responses ($n = 12$), illustrating how vibrotactile stimulation shifts the perceived location of skin-stretch beyond the physically actuated site.

Table 2: Mean ($\pm$ SD) of Onset Location of skin-stretch ($n = 12$).

	Condition Left (Mean $\pm$ SD)	Right (Mean $\pm$ SD)
C1	0.38 $\pm$ 0.11	0.38 $\pm$ 0.11
C2	0.56 $\pm$ 0.12	0.34 $\pm$ 0.14
C3	0.69 $\pm$ 0.19	0.44 $\pm$ 0.08
C4	0.37 $\pm$ 0.14	0.56 $\pm$ 0.11
C5	0.40 $\pm$ 0.08	0.70 $\pm$ 0.21
C6	0.77 $\pm$ 0.16	0.78 $\pm$ 0.16
C7	0.70 $\pm$ 0.13	0.56 $\pm$ 0.14
C8	0.67 $\pm$ 0.17	0.40 $\pm$ 0.08

Table 3: Significant post-hoc comparisons (paired t -tests with Holm correction). Only pairs with $p_{\text{Holm}} < .05$ are shown.

Left			Right		
Pair	p_{Holm}	d_z	Pair	p_{Holm}	d_z
C1-C2	.001	-2.01	C1-C4 <	.001	-2.36
C1-C3	.003	-1.60	C1-C5	.004	-1.54
C1-C6	.001	-2.02	C1-C6 <	.001	-2.18
C1-C7 <	.001	-2.18	C1-C7	.024	-1.19
C1-C8	.006	-1.43	C2-C4 <	.001	-2.29
C2-C3	.027	-1.12	C2-C5 <	.001	-2.09
C2-C4	.004	1.53	C2-C6 <	.001	-2.44
C2-C5	.002	1.71	C2-C7	.002	-1.68
C2-C6	.003	-1.64	C3-C4	.036	-1.08
C2-C7	.001	-1.84	C3-C5	.008	-1.39
C3-C4	.002	1.77	C3-C6 <	.001	-2.42
C3-C5	.006	1.44	C3-C7	.050	-0.99
C4-C6 <	.001	-2.16	C4-C5	.050	-1.01
C4-C7 <	.001	-2.65	C4-C6	.001	-1.77
C4-C8	.003	-1.58	C4-C8	.031	1.12
C5-C6	.001	-1.98	C5-C7	.018	1.25
C5-C7 <	.001	-2.07	C5-C8	.007	1.44
C5-C8	.007	-1.37	C6-C7 <	.001	2.04
			C6-C8 <	.001	2.42
			C7-C8	.029	1.14

was concluded when the 90% confidence interval of the mean difference was entirely contained within the interval $[-\Delta, +\Delta]$.

In conclusion, simultaneous vibrotactile stimulation modulates the perceived location of skin-stretch sensation beyond the physically actuated site. Furthermore, the perceived skin-stretch location did not change even when the vibrators were driven in a sparse configuration, that is, when the skin-stretch location and the vibrator positions were spatially separated (table 4, Left, C3 and C8).

4.2 Symmetry between the Left and Right Sides

To examine responses to laterally symmetric stimuli, Condition 1 was defined as a baseline, and contrastive vibratory stimuli were applied in Conditions 2 through 6. Focusing on Conditions 2–6, equivalence between left and right sides and across condition pairings was assessed using paired Two One-Sided Tests (TOST). The equivalence margin was set to $\Delta = 0.1$. The significance level was set at $\alpha = .05$, and equivalence was concluded when the 90% confidence interval of the mean difference was entirely contained within the interval $[-\Delta, +\Delta]$. The tested pairings and their results are summarized in Table 5. Equivalence was established for all pairings except 5L–3R.

Table 4: Equivalence test results (TOST, $\Delta = 0.1$). Only pairs judged equivalent are shown.

Left			Right		
Pair	MeanDiff	p_{TOST}	Pair	MeanDiff	p_{TOST}
C1–C4	0.005	.007	C1–C2	0.038	.018
C1–C5	-0.025	< .001	C3–C8	0.020	< .001
C3–C7	-0.040	.018	C4–C7	0.000	.003
C3–C8	0.013	.003			
C4–C5	-0.030	.042			
C6–C7	0.038	.008			

Table 5: Validation of equivalence between symmetrical tactile stimuli ($\Delta = 0.1$)

Pair	MeanDiff	p_{TOST}
1L–1R	0.003	< 0.001
2L–4R	0.008	< 0.001
3L–5R	-0.015	< 0.001
4L–2R	0.005	0.017
5L–3R	-0.038	0.034
6L–6R	-0.013	< 0.001

4.3 Confidence Level

Figure 4 summarizes the confidence ratings for each condition except C7. Since all factors were within-subject variables, the normality of the data for each condition was first examined using the Shapiro–Wilk test. As a result, normality was rejected for all conditions (all $p < .05$). Therefore, non-parametric statistical methods were employed for the subsequent analyses.

To examine overall differences among conditions, a Friedman test was conducted. The results revealed a significant main effect of condition ($\chi^2(7) = 33.61$, $p < .001$), indicating that perceived confidence differed significantly across stimulation conditions. As an effect size corresponding to the Friedman test, Kendall’s coefficient of concordance was calculated, yielding $W = 0.40$, which corresponds to a medium effect.

Given the significant overall effect, post-hoc pairwise comparisons between conditions were performed using Wilcoxon signed-rank tests. To control for Type I error due to multiple comparisons, Holm correction was applied to the p -values. In C8, where the vibrators were positioned at a distance from the skin-stretch location, the mean confidence level was lower than that in the other conditions, but the difference was not statistically significant.

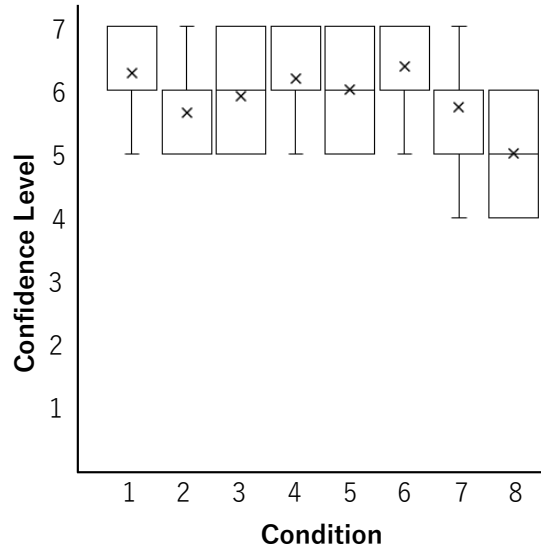


Fig. 4: Confidence ratings for perceived skin-stretch location across conditions.

5 Discussion

5.1 Summary of Experimental Findings

From Table 2, it can be observed that the perceived skin-stretch location changes depending on the vibration conditions, and this effect is statistically supported by the results of the analysis of variance. While one possible interpretation is that participants' reports were influenced by response bias or increased spatial uncertainty, the observed shifts were systematic, directional, and condition-dependent. Moreover, the shifts exhibited left-right symmetry under symmetric stimulation patterns, suggesting a perceptual integration process rather than random reporting variability. Moreover, Despite the modest sample size, the within-subject design and consistently large effect sizes observed across conditions indicate that the reported effects are robust.

Focusing on the results from C1 to C3, these conditions differ in terms of which vibrator located in the vicinity on the left side of the skin-stretch point was activated. The perceived skin-stretch location expands outward in the order of C1, C2, and C3. As shown in Table 3, this difference is statistically supported. In contrast, the perceived skin-stretch location on the right side does not change substantially under these conditions. Although C3 was not regarded as statistically equivalent, C1 and C2 were shown to be statistically equivalent, and a similar tendency was observed when comparing C1, C4, and C5. Furthermore, in the evaluation of left-right symmetry, changes in the perceived skin-stretch location induced by symmetrically arranged vibrators were also symmetric. The

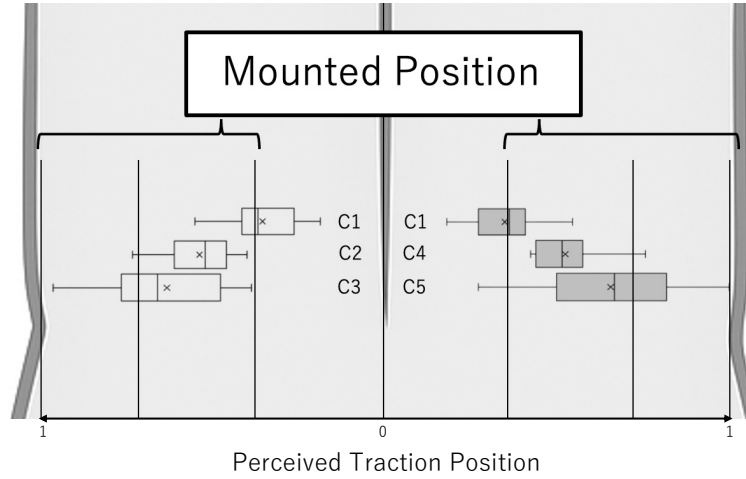


Fig. 5: Relationship between physical device placement and perceived skin-stretch positions.

obtained P_{TOST} values were smaller than those for unilateral skin-stretch comparisons across different conditions discussed in the previous subsection(4, 5). These results indicate that the extension of the skin-stretch sensation exhibits strong left-right symmetry.

Focusing on the perceived right-side skin-stretch location in conditions C1, C2, C3, and C8, in all of these conditions, the right-side vibrator was not activated, while the combinations of activated left-side vibrators differed. Although no statistically significant differences were observed, the perceived skinstretch locations obtained in C3 and C8 were larger than those in C1 and C2. Notably, in C3 and C8, the perceived left-side skin-stretch locations were significantly larger than those in C1 and C2. From these results, the following conclusion can be drawn. By activating vibrators in the vicinity of one side of the skin-stretch point, it is possible to shift the perceived skin-stretch location on that side alone, and the magnitude of this shift can be controlled by adjusting the vibrator arrangement. However, when inducing a large shift in the perceived skin-stretch location, there is a possibility that the opposite side may also be affected.

As a noteworthy result, equivalence was confirmed between the perceived left-side skin-stretch locations in C3 and C8. This indicates that vibrations applied at locations beyond the two-point discrimination threshold can still shift the perceived skin-stretch point. On the other hand, although no significant difference was observed, Figure 4 shows that the confidence level in C8 was lower, suggesting that, in terms of user experience, a large spatial separation between the vibrator and the perceived skin-traction location may act as noise.

The normalized positions of the mounted device parts varied depending on the participant's waist size, but they were generally around 0.4, 0.8, and 1.0. Fig-

ure 5 summarizes the attachment positions of the device parts, the skin-stretch positions perceived under C1, and the shifted skin-stretch positions under C2 through C5. Under C1, the perceived skin-stretch position was generally consistent with the actual attachment position of the device parts. This indicates that most participants were able to accurately identify the location of a single stimulus produced by skin-traction. When comparing the perceived skin-stretch positions obtained under C2 and C3 with the device part attachment positions, a discrepancy can be observed. This suggests that driving the vibrators can induce a shift in the perceived skin-stretch position; however, the perceived position generally lies between the true skin-traction location and the vibrator. These results suggest that the observed perceptual shift is more consistent with a process of tactile cue integration rather than a complete relocation of the perceived stimulus. In most cases, the perceived location lies between the physical skin-traction point and the vibrotactile actuator, indicating an interpolative mechanism that integrates heterogeneous tactile signals into a unified spatial estimate.

5.2 Design Implications

The results of this study suggest the feasibility and potential effectiveness of a design approach for wearable haptic displays with a limited number of actuators, in which vibrotactile stimuli are used as cues to guide the spatial estimation of skin-stretch sensations. Based on these findings, the following design guidelines are summarized below:

- The perceived location of a skin-stretch stimulus is not determined solely by the physical location of the skin-traction unit, but can be reconstructed outward through surrounding vibrotactile stimulation. Therefore, vibration should not be treated as a simple addition of intensity; instead, it should be strategically placed as an integrative cue that enables users to infer the origin and spatial extent of the skin-stretch sensation. In particular, adding vibrotactile cues outside the physical skin-traction location is effective.
- On the other hand, when the magnitude of the perceptual shift is large, changes may also appear in responses on the contralateral side. This introduces a trade-off between *local control* and *global integration* (i.e., bilateral propagation of perception). Accordingly, when a design aims to strongly emphasize motion on only one side, it is necessary to suppress vibrations on the opposite side or to gradually increase vibration intensity on the same side, so that perception is not unintentionally drawn into a global response.
- Although the perceived skin-stretch location shifted outward from the physical position of the skin-traction unit, it tended to remain within the region between the skin-traction unit and the vibrotactile actuator. This suggests that, rather than placing vibrotactile actuators directly at the target perceptual location, it is effective to position them as guiding points that lead perception from the skin-traction location toward the target. In other words, the relative arrangement of the skin-traction unit and vibrotactile actuators may allow the landing point of perception to be adjusted in an interpolative manner.

- This method shows high design reproducibility when symmetric left–right patterns are used for stimulation on the back. As a result, it is well suited to applications such as bodily extension, where plausibility (e.g., left–right symmetry) is more important than the precision of sensory localization itself.

From a practical perspective, the findings of this study suggest that the number of skin-stretch actuators required in wearable systems may be reduced by leveraging vibrotactile cues to modulate perceived location. For example, a small number of skin-stretch units combined with distributed vibrotactile actuators could create the impression of spatially extended deformation across the torso without requiring dense mechanical actuation. This approach is particularly beneficial for wearable devices, where constraints on weight, power consumption, and form factor limit the number of deployable actuators, and it may enable scalable haptic designs for applications such as navigation guidance, motion assistance, and body extension interfaces.

5.3 Limitation and Future work

This study has several limitations. First, the present evaluation relied on subjective reports of perceived skin-stretch location using schematic body maps. While this approach is appropriate given the coarse body representation of the back, it may limit spatial precision and generalizability. In addition, the study focused exclusively on the back, and the extent to which similar perceptual integration effects occur on body parts with higher tactile acuity remains to be investigated. Another possible interpretation of the present results is that participants may have partially relied on the perceived location of the vibrotactile stimulus when reporting the location of skin stretch, particularly when the vibrotactile cue was perceptually salient. Therefore, the present findings may reflect not only a shift in the perceived location of the skin-stretch itself, but also an integration process between traction and vibrotactile cues. Future work should disentangle these factors by independently measuring the perceived location of skin stretch and vibration, and by systematically manipulating the relative intensity and duration of the two cues.

Second, the stimulus conditions were restricted to specific parameter settings, including a fixed vibration frequency, duration, and skin-traction velocity and displacement. As a result, the explored stimulus parameter space was limited. In particular, vibration intensity, onset timing, and the temporal relationship between vibration and skin-traction may substantially affect perceptual integration.

Furthermore, in this study, the device was designed to be bilaterally symmetrical and focused on the medial–lateral axis in order to ensure that the two skin-traction points possessed physically and perceptually equivalent properties. By also investigating whether similar perceptual modulation occurs along the cranial–caudal axis, it is expected that a deeper understanding of the generalizability of this method can be achieved.

In future work, it will be necessary to introduce more robust evaluation methods, such as relative judgment and repetitive tasks, and to systematically manipulate temporal factors, intensity parameters, and traction direction in order to clarify the conditions under which the perceived location of skin-traction shifts in a stable manner.

Furthermore, the number of participants in this experiment was limited, and each condition was tested only once. This may reduce statistical power and increase variability in the results. Future work should increase the number of trials per condition and expand the participant pool to improve the robustness and generalizability of the findings. In addition, the results of the equivalence testing depend on the choice of the equivalence margin ($\Delta = 0.1$), which was determined based on actuator spacing. However, different choices of Δ could lead to different conclusions regarding equivalence.

Future work should evaluate robustness across multiple equivalence margins and explore perceptually grounded thresholds.

6 Conclusion

In this study, we proposed a device that alters the perceived location of skin-stretch sensation using vibrotactile stimulation, with a focus on the tactile characteristics of the back. The device is a wearable system worn on the back, and experimental results confirmed that placing vibrators adjacent to the area where the skin is stretched can shift the perceived location of the skin-stretch sensation toward the direction of vibration. Furthermore, based on the obtained results, we discussed design insights for wearable skin-traction devices with significant mechanical constraints, focusing on how human perceptual characteristics can be leveraged in device design.

Acknowledgments. This work was supported by JSPS Fellows (Grant Number JP25KJ1922) and in part by JSPS KAKENHI (No. JP24H00892).

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Abaira, V.E., Ginty, D.D.: The sensory neurons of touch. *neuron* **79**(4), 618–639 (2013)
2. Campanelli, A., Tiboni, M., Verité, F., Saudrais, C., Mick, S., Jarrassé, N.: Innovative multi vibrotactile-skin stretch (muwiss) haptic device for sensory motor feedback from a robotic prosthetic hand. *Mechatronics* **99**, 103161 (2024)
3. Chossat, J.B., Chen, D.K., Park, Y.L., Shull, P.B.: Soft wearable skin-stretch device for haptic feedback using twisted and coiled polymer actuators. *IEEE transactions on haptics* **12**(4), 521–532 (2019)
4. Collins, C.C.: Tactile television-mechanical and electrical image projection. *IEEE Transactions on man-machine systems* **11**(1), 65–71 (2007)

5. Geldard, F.A., Sherrick, C.E.: The cutaneous "rabbit": a perceptual illusion. *Science* **178**(4057), 178–179 (1972)
6. Gleeson, B.T., Horschel, S.K., Provancher, W.R.: Perception of direction for applied tangential skin displacement: Effects of speed, displacement, and repetition. *IEEE transactions on haptics* **3**(3), 177–188 (2010)
7. Green, B.G.: Referred thermal sensations: warmth versus cold. *Sensory processes* **2**(3), 220–230 (1978)
8. Jones, L.A., Sarter, N.B.: Tactile displays: Guidance for their design and application. *Human factors* **50**(1), 90–111 (2008)
9. Konishi, Y., Hanamitsu, N., Outram, B., Minamizawa, K., Mizuguchi, T., Sato, A.: Synesthesia suit: the full body immersive experience. In: *ACM SIGGRAPH 2016 VR Village*, pp. 1–1 (2016)
10. Li, J., Zhang, Z., Yan, Z., Zhao, Y., Peng, H.: Comparing vibrotactile and skin-stretch haptic feedback for conveying spatial information of virtual objects to blind vr users. *2025 IEEE Conference Virtual Reality and 3D User Interfaces (VR)* pp. 686–696 (2024). <https://doi.org/10.1109/VR59515.2025.00091>
11. Lieberman, J., Breazeal, C.: Development of a wearable vibrotactile feedback suit for accelerated human motor learning. In: *Proceedings 2007 IEEE International Conference on Robotics and Automation*. pp. 4001–4006. IEEE (2007)
12. Maisto, M., Pacchierotti, C., Chinello, F., Salvietti, G., De Luca, A., Prattichizzo, D.: Evaluation of wearable haptic systems for the fingers in augmented reality applications. *IEEE transactions on haptics* **10**(4), 511–522 (2017)
13. Miyakami, M., Takahashi, A., Kajimoto, H.: Head rotation and illusory force sensation by lateral skin stretch on the face. *Frontiers in Virtual Reality* **3**, 930848 (2022)
14. Park, H., Pei, J., Shi, M., Xu, Q., Fan, J.: Designing wearable computing devices for improved comfort and user acceptance. *Ergonomics* **62**, 1474 – 1484 (2019). <https://doi.org/10.1080/00140139.2019.1657184>
15. Piao, J., Kim, M., Kim, J., Kim, C., Han, S.J., Back, I., sung Koh, J., Koo, S.: Development of a comfort suit-type soft-wearable robot with flexible artificial muscles for walking assistance. *Scientific Reports* **13** (2023). <https://doi.org/10.1038/s41598-023-32117-2>
16. Takashita, S., Steimle, J., Inami, M.: Imaginary joint: Proprioceptive feedback for virtual body extensions via skin stretch pp. 63:1–63:15 (2025). <https://doi.org/10.1145/3746059.3747800>
17. Tanabe, T., Endo, H., Ino, S.: Effects of asymmetric vibration frequency on pulling illusions. *Sensors* **20**(24), 7086 (2020)
18. Watanabe, R., Kajimoto, H.: Pressure sensation elicited by rapid temperature changes. In: *Haptic Interaction: Perception, Devices and Applications*, pp. 41–44. Springer (2015)
19. Yang, T.H., Kim, J.R., Jin, H., Gil, H., Koo, J.H., Kim, H.J.: Recent advances and opportunities of active materials for haptic technologies in virtual and augmented reality. *Advanced Functional Materials* **31**(39), 2008831 (2021)